TRAUMA · NEET-SS / FET / INI-SS

Proximal humerus and shoulder girdle

1. Proximal humerus fractures

Anatomy

- Four parts (**Codman**): **head** (articular segment), **greater tuberosity** (supraspinatus, infraspinatus, teres minor), **lesser tuberosity** (subscapularis), **shaft**.
- **Head blood supply:** the **posterior humeral circumflex artery** supplies most of the head (Hettrich, 2010; about 64%); the **anterior humeral circumflex artery** and its **arcuate artery (of Laing)**, entering at the bicipital groove, was classically considered the main supply.
- **Hertel's predictors of head ischaemia: metaphyseal extension under 8 mm** attached to the head (posteromedial calcar), **disrupted medial hinge (over 2 mm displacement)**, and the **anatomical neck fracture** pattern. Head ischaemia does not always mean AVN.
- **Deforming forces:** greater tuberosity pulled **superiorly and posteriorly** (supraspinatus/infraspinatus); lesser tuberosity **medially** (subscapularis); shaft **medially and anteriorly** (pectoralis major).
- **Axillary nerve** runs about **5-7 cm below the acromion** (deltoid-splitting approach danger zone); the **arcuate artery** and **cephalic vein** in the deltopectoral approach.

Neer classification

- A "part" is displaced if **over 1 cm displacement or over 45 degrees angulation** (for the greater tuberosity, **over 5 mm displacement** is significant in young/overhead patients, as superior displacement causes impingement).
- One-part (undisplaced, about 80%), two-part (surgical neck commonest two-part; greater tuberosity; anatomical neck rare),

three-part (surgical neck + one tuberosity), four-part (both tuberosities + surgical neck), **fracture-dislocations** and **head-splitting** fractures.

- **Valgus-impacted four-part fracture:** better prognosis (medial periosteal hinge preserved, lower AVN).
- **AO/OTA 11** classification also used.

Management

- **Non-operative** (sling, early pendulum exercises at about 2 weeks): most fractures, especially **one-part** and in the **elderly**. The **ProFHER trial** (2015, elderly displaced surgical-neck-involving fractures): **no benefit of surgery over non-operative care** at 2 and 5 years.
- **Isolated greater tuberosity fracture: fix if displaced over 5 mm** (screws, suture-bridge/tension band) in active patients.
- **Two/three-part fractures in active patients: proximal humeral locking plate** (deltopectoral or deltoid-split approach). **Key technical points:** restore the **medial calcar support** (an inferomedial "calcar screw"; medial comminution predicts varus collapse), avoid **plate placed too high** (subacromial impingement; plate about 8 mm below the top of the greater tuberosity), **avoid screw cut-out** (commonest complication: primary or secondary intra-articular screw penetration with varus collapse), use **tuberosity sutures** through the cuff. Fibular strut allograft in the canal for medial support.
- **Intramedullary nails** for two-part surgical neck fractures (straight nails reduce cuff damage).
- **Four-part fractures, head-split, fracture-dislocations in the elderly:**
  - **Reverse total shoulder arthroplasty (RTSA)** is now preferred in elderly patients (over about 70): **more predictable function**, does not depend on tuberosity healing (though tuberosity repair improves rotation).

  - **Hemiarthroplasty** in younger patients where tuberosity healing is likely: outcome **depends on tuberosity healing** (correct height and version: top of the head about 5-8 mm above the greater tuberosity, retroversion about 20-30 degrees).
  - **ORIF** in young patients (attempt to preserve the head; AVN may still occur).
- **Complications: AVN** (four-part and head-split, about 10-30%+), malunion (varus, tuberosity), non-union (surgical neck), **screw cut-out**, stiffness, **axillary nerve injury**, infection.

2. Shoulder dislocation (acute, trauma aspect)

- **Anterior (about 95-97%):** abduction-external rotation; associated **Bankart lesion** (anteroinferior labrum/capsule), **Hill-Sachs** (posterosuperolateral head impaction), **greater tuberosity fracture** (in older patients; often reduces with the head), **axillary nerve injury** (commonest nerve), **rotator cuff tear** in patients over 40 (**test cuff strength after reduction; MRI/ultrasound if weak**).
- **Recurrence after first dislocation** is inversely related to age: **up to about 90% under 20 years**, much lower over 40. Arthroscopic Bankart repair after first dislocation lowers recurrence in young athletes.
- **Posterior: seizures, electrocution**, **lightbulb sign**, **trough (reverse Hill-Sachs/McLaughlin) sign**, inability to externally rotate; **often missed** (always get an **axillary lateral** view). Reverse Hill-Sachs over about 25% of the head: McLaughlin procedure (subscapularis/lesser tuberosity transfer); over about 50%: arthroplasty/allograft.
- **Luxatio erecta** (inferior): arm locked in abduction; neurovascular injury common.

3. Clavicle fractures

- Commonest fracture of the shoulder girdle; **middle third (about 80%)** (Allman group I), lateral third (about 15%, group II), medial

third (about 5%, group III).

- **Midshaft:** most heal non-operatively. **Displaced/shortened midshaft fractures** (100% displacement, **shortening over about 2 cm**, comminution) have higher **non-union (about 15%)** and **symptomatic malunion** rates without surgery (**COTS trial**, 2007: **plate fixation lowered non-union and improved function** in displaced fractures; later trials show smaller differences). **Absolute indications: open fracture, skin tenting with impending perforation, neurovascular injury**, floating shoulder (relative), polytrauma.
- **Fixation:** superior or anteroinferior plate (anteroinferior: less prominence, longer screws, less subclavian vessel risk) or intramedullary devices.
- **Lateral third (Neer): Type I** (lateral to the coracoclavicular ligaments, stable), **Type II** (medial to CC ligaments; the medial fragment detaches from the CC ligaments, **high non-union (about 30%)**: **fix** with hook plate, lateral locking plate or CC suspension), **Type III** (intra-articular into AC joint; late AC arthritis), IV (children, periosteal sleeve), V (comminuted).
- **Medial clavicle / sternoclavicular:** posterior SC dislocation endangers great vessels and trachea (**CT; reduce in theatre with a thoracic surgeon available**); medial physis closes at about 23-25 years (physeal injuries mimic dislocations).

4. Acromioclavicular (AC) joint injuries

- **Rockwood classification:**

Type	AC ligament	CC ligament	Displacement	Treatment
I	Sprained	Intact	None	Sling
II	Torn	Sprained	Up to 25% (subluxed)	Sling

Type	AC ligament	CC ligament	Displacement	Treatment
III	Torn	Torn	**25-100%** CC distance increase	**Non-operative first** in most; surgery for heavy labourers/overhead athletes or failed rehab
IV	Torn	Torn	**Clavicle posterior** through trapezius (axillary view)	Surgery
V	Torn	Torn	**100-300%** (deltotrapezial fascia stripped)	Surgery
VI	Torn	Torn	Clavicle **inferior** (subacromial or subcoracoid)	Surgery (rare)

- **Zanca view** (10-15 degrees cephalic tilt) for the AC joint; **normal CC distance about 11-13 mm**.
- Surgery: anatomical CC ligament reconstruction (graft), suture-button suspension, hook plate (acute), **Weaver-Dunn** (coracoacromial ligament transfer, historical).

5. Scapula fractures

- High-energy injuries; **associated injuries in about 80-90%** (rib fractures, pneumothorax, pulmonary contusion, head and spine injury).
- **Body fractures** (commonest): mostly non-operative. Operate for significant **medialisation of the glenoid (over 20 mm)**, **angulation over 45 degrees**, **glenopolar angle under 22 degrees**.
- **Glenoid fractures (Ideberg):** fix if **articular step-off over 4-5 mm**, or rim fractures causing **instability (over about 20-25% of the**

glenoid).

- **Floating shoulder:** ipsilateral clavicle and glenoid neck fractures (superior shoulder suspensory complex, **Goss**: double disruption of the SSSC). Fix one or both when displaced.
- **Scapulothoracic dissociation:** lateral displacement of the scapula (**scapular index**), often with **brachial plexus avulsion and subclavian/axillary artery injury**; high mortality; flail arm, often amputation.

6. Humeral shaft fractures

- **Holstein-Lewis fracture:** spiral fracture of the distal third with **radial nerve** entrapment risk.
- **Non-operative (Sarmiento functional brace)** is standard: acceptable alignment (Klenerman) **under 20 degrees anterior angulation, under 30 degrees varus, under 3 cm shortening** (Campbell's: under 3 cm shortening, under 20 degrees angulation, under 30 degrees rotation). **FISH trial** (2021) and others: surgery gives faster early recovery but similar long-term function; non-union about 10-15% with bracing (proximal third and transverse patterns more).
- **Operative indications: open fracture**, **vascular injury**, **floating elbow**, polytrauma (to allow crutch use), **pathological fracture**, bilateral fractures, failed bracing, segmental fractures, intra-articular extension, brachial plexus injury.
- **Plating** (anterolateral, posterior or MIPO) is preferred over **antegrade nailing** (nailing causes more **shoulder pain** and reoperation; useful in pathological fractures).
- **Radial nerve palsy with a closed fracture** (about **12%** overall in a large meta-analysis; commonest with middle and middle-distal third, transverse and spiral fractures): mostly **neurapraxia**: **observe** (about 70-90% recover spontaneously; wait about 3-4 months, then EMG; explore if no recovery). **Explore early** for **open fractures**

(nerve may be lacerated) and palsy with an **open or high-energy fracture needing fixation**. **Palsy after closed manipulation** is debated (traditionally explore). Tendon transfers (pronator teres to ECRB, FCR/FCU to EDC, palmaris longus to EPL) if no recovery.

Exam pearl: **Neer part = over 1 cm or 45 degrees (greater tuberosity over 5 mm). ProFHER: elderly displaced fractures do as well without surgery. Reverse arthroplasty is preferred for four-part fractures in the elderly; hemiarthroplasty depends on tuberosity healing. Medial calcar support prevents varus collapse after locking plates.**

Exam pearl: **Posterior shoulder dislocation (seizure, electrocution) = lightbulb sign, missed without an axillary view. Neer II lateral clavicle fracture has a high non-union rate: fix. Rockwood III AC injury: non-operative first. Radial nerve palsy with closed humeral shaft fracture: observe.**

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.